

Bureau of Indian Standards

Manak Bhawan, 9, Bahadur Shah Zafar Marg
New Delhi - 110002

IS 269:2015

2015

ORDINARY PORTLAND CEMENT, 33 GRADE

SPECIFICATION

ICS 91.100.10
BIS 2015

Indian Standard

1 SCOPE

This standard covers the manufacture and properties of ordinary portland cement of 33 grade, used for general construction purposes including plain and reinforced concrete, masonry mortar, and plastering.

2 COMPOSITION

The cement shall be produced by intimately mixing together calcareous and argillaceous and/or other silica, alumina or iron oxide-bearing materials, burning them at a clinkering temperature, and grinding the resultant clinker with a small quantity of gypsum.

- Tricalcium Silicate (C3S): 45-65%
- Dicalcium Silicate (C2S): 10-30%
- Tricalcium Aluminate (C3A): 0-8%
- Tetracalcium Aluminoferrite (C4AF): 6-20%

3 PHYSICAL REQUIREMENTS

Property	Requirement	Test Method
Fineness (Blaine)	Min 225 m ² /kg	IS 4031 (Part 2)
Soundness (Le Chatelier)	Max 10 mm	IS 4031 (Part 3)
Setting Time - Initial	Min 30 minutes	IS 4031 (Part 5)
Setting Time - Final	Max 600 minutes	IS 4031 (Part 5)
Compressive Strength - 3 days	Min 10 MPa	IS 4031 (Part 6)

Compressive Strength - 7 days	Min 16 MPa	IS 4031 (Part 6)
Compressive Strength - 28 days	Min 33 MPa	IS 4031 (Part 6)

4 CHEMICAL REQUIREMENTS

Oxide	Percentage by Mass
Magnesia (MgO)	Max 6.0%
Sulfur Trioxide (SO ₃)	Max 3.5%
Chloride (as Cl ⁻)	Max 0.10%
Loss on Ignition	Max 5.0%
Insoluble Residue	Max 4.0%
Lime Saturation Factor	0.66-1.02

5 MARKING

Each bag or container shall be distinctly marked with the following: (a) Name and trade mark, (b) IS 269:2015, (c) Net quantity, (d) Date and time of packing, (e) Lot number, (f) Grade designation: 33.